DocGuard: Forensic Preprocessor

Module 1 Technical & Pedagogical Guide — Understanding the Signal Extraction Engine

1. What is Module 1 Doing? (The Objective in Simple Language)

Imagine someone hands you a printed medical receipt where the total amount has been altered from **INR 4,000** to **INR 14,000** using Photoshop or an AI editing tool. To the human eye, the numbers look completely normal. Standard Optical Character Recognition (OCR) simply reads 'INR 14,000'. And because an auditor rarely has the clinic's original reference document, standard hash checks or visual inspections fail.

Module 1 is the digital forensic investigator that examines the physical physics underneath the surface pixels.

Its objective is **Forensic Signal Extraction**: before passing any document to deep neural networks, Module 1 extracts two hidden physical signatures that humans cannot see, but which prove whether an image has been altered:

• **1. Compression Equilibrium Discrepancies (ELA):** Inspects whether different areas of the document have been compressed at different times (exposing altered numbers, modified dates, or inserted text).

• **2. Paper Grain & Noise Discontinuities:** Analyzes the microscopic physical paper texture and sensor noise across the page to expose unnaturally smooth ('sterile') patches left by AI inpainting or digital erasing.

All calculations execute **100% in system memory (RAM)** via virtual streams (io.BytesIO), eliminating disk I/O latency and preventing Windows file-locking crashes.

2. How Does Module 1 Achieve This? (Three-Stage Workflow)

Stage	Core Operation	Forensic Purpose & Physical Output
Stage 1: Standardization	Converts input to RGB, applies EXIF transposition, and composites transparency over pure white paper.	Guarantees clean, upright 3-channel images without black-border transparency artifacts.
Stage 2: In-Memory ELA	Re-compresses image in RAM at Quality 95 and computes absolute difference: |I_orig - I_resaved|.	Untouched text remains dark; newly modified numbers glow bright white due to first-time lossy quantization.
Stage 3: Noise Residuals	Applies a 3x3 non-linear median filter and subtracts smoothed image: |I - I_smooth|.	Isolates paper fiber grain and sensor noise. Exposes digitally erased/inpainted patches that lack natural texture.

3. Comprehensive Glossary of Technical & ML Concepts

Below is an explanation of every technical term and machine learning concept used in Module 1.

A. Digital Image Fundamentals & Color Spaces

• **Pixels & RGB Color Space:** A digital image is a 2D matrix of colored dots (pixels). Standard images use RGB (Red, Green, Blue), where each pixel has 3 numbers from 0 to 255. (255, 255, 255) is pure white paper, and (0, 0, 0) is black printed ink.

• **The Alpha Channel & 'Black Border' Trap:** In PNG scans or digital stamps, a 4th channel (Alpha) controls transparency. If an image is naively converted to RGB, transparent pixels automatically turn pitch-black (0, 0, 0). On white paper, this creates an extreme artificial gradient jump [255 -> 0] that tricks neural networks into false alarms. Module 1 solves this by **alpha-compositing onto a white canvas**, making transparency appear as authentic white paper.

• **EXIF Orientation & Matrix Transposition:** Mobile phone cameras record orientation metadata (e.g. 'Rotate 90 deg') without physically rotating the pixel grid. Module 1 reads this EXIF tag and transposes the matrix upright so text is horizontal.

B. Compression Physics & Error Level Analysis (ELA)

• **Lossy vs. Lossless Compression:** PNG and TIFF are lossless (zero data lost). JPEG is lossy—it permanently discards fine details the human eye cannot easily notice to make files smaller.

• **Discrete Cosine Transform (DCT):** JPEG splits images into 8x8 pixel blocks and converts spatial pixels into 64 frequency waves. Low frequencies represent paper background; high frequencies represent sharp character edges.

• **Quantization & Quantization Matrices:** The lossy step in JPEG. Frequency coefficients are divided by a Quantization Table and rounded to whole integers. Fractional remainders are discarded forever.

• **Quantization Equilibrium:** When an unedited JPEG is re-saved at the same quality level, its pixels have already been divided by that quantization table. Dividing again causes near-zero rounding error (Delta ~ 0). This stability is called Quantization Equilibrium.

• **The ELA Tampering Anomaly:** When an attacker modifies numbers (e.g. altering INR 4,000 to INR 14,000), those altered digits are saved for the *very first time*. When Module 1 compresses the whole document at Quality 95 and computes the absolute difference, untouched text in equilibrium stays dark, while the newly inserted digits undergo active lossy quantization and **glow bright white**.

C. Spatial Noise, Filtering & AI Inpainting

• **Sensor Noise & Paper Grain:** Authentic scans possess microscopic high-frequency texture caused by camera photon shot noise, thermal noise, and the physical cellulose grain of authentic wood fibers in paper.

• **Non-Linear Median Filtering vs. Linear Blur:** Standard linear Gaussian blur averages neighboring pixels, turning black text and white paper into blurry gray halos. A 3x3 median filter sorts neighborhood pixels and picks the middle value. It is *edge-preserving*: it strips isolated noise dots while keeping character edges crisp.

• **Spatial Noise Residual:** Subtracting the median-smoothed image from the original isolates the high-pass texture: R(x, y) = |I - I_smooth|.

• **The 'Sterile Patch' Anomaly:** When an attacker uses Adobe Photoshop 'Content-Aware Fill' or Generative AI (Stable Diffusion) to erase a stamp or text, the AI algorithm paints over the area smoothly. **The AI cannot recreate the microscopic physical noise of the scanner or paper.** In the noise residual map, authentic paper has a healthy grain variance, while the erased area has zero noise (mean = 0.0), exposing the alteration immediately.

D. Machine Learning & Deep Learning Terminology

• **Why Raw RGB Confuses Standard CNNs:** If a Convolutional Neural Network (CNN) is trained only on raw RGB images, it often memorizes document layouts and company logos rather than forensic tampering cues. It may flag a genuine document simply because it has never seen that clinic's logo.

• **Feature Decoupling & Inductive Bias:** By providing pre-computed ELA maps and noise residuals, we decouple the physics of tampering from the document's visual text. Stream 1 (EfficientNet-B0) is forced to specialize in compression artifacts, while Stream 2 (ResNet-50) looks at font geometry.

• **Out-of-Distribution (OOD) Robustness:** Because Module 1 is based on physical laws (JPEG quantization math and sensor noise variance) rather than memorized visual templates, it can detect alterations produced by brand-new generative AI tools that did not exist during training.

E. In-Memory Streaming & Operating System Architecture

• **Virtual RAM Streams (io.BytesIO):** Standard programs write temporary images to the hard drive (C:\temp\image.jpg). io.BytesIO creates a virtual file in RAM. Python interacts with it like a real file, but it runs at sub-millisecond RAM speeds.

• **Windows File Locking (WinError 32):** On Windows, opening a disk file locks it. If another thread or antivirus tries to access it, Windows crashes with WinError 32. Executing 100% in RAM via io.BytesIO completely eliminates disk latency and file-locking errors.

4. Summary: How Module 1 Signals Feed the Neural Networks

Forensic Signal	Physical Principle	What Fraud Vector It Detects	Target Consumer
Standardized RGB	White alpha-compositing & EXIF transposition	Eliminates false border spikes and sideways scans	Stream 2 (ResNet-50) & Visual UI
In-Memory ELA Map	JPEG Quantization Equilibrium delta	Altered amounts (INR 4k -> 14k), modified dates, inserted text	Stream 1 (EfficientNet-B0)
Noise Residual Matrix	Non-linear 3x3 median filter subtraction	Erased stamps, removed text, AI inpainting (sterile patches)	Decision Fusion Brain & Red-Box Localizer